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COI Name (Institution):	Fluckey, James Ph.D. (Texas A&M University)		
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Task Description:	A critical limitation to astronaut performance during spaceflight missions including those to the International Space Station and Mars is the impaired contractile ability of skeletal muscles. Skeletal muscle is a highly specialized tissue that rapidly adapts to dynamic changes in mechanical loading by modifying its ultrastructure and mass. Detection and transmittance of loading are vital to muscle integrity and protein turnover, with a nexus associated with the subsarcolemmal cytoskeleton that includes the dystrophin-glycoprotein complex (DGC) and caveolae. This interface between myocytes and the extracellular matrix confers mechanical integrity and activates cell signaling pathways that regulate protein turnover, mitochondrial, muscle fiber cross-sectional area, and fiber phenotype. Mechanical unloading experienced in the microgravity of spaceflight elicits large decrements in force generating capacity via reduction in muscle fiber cross sectional area or atrophy. While unloading-induced atrophy is a function of reduced protein synthesis coupled with increased protein degradation, activation of the transcription factors FoxO3a and nuclear factor-kappaB have drawn significant attention as mediators. Although the upstream triggers are not fully understood, translocation of neuronal nitric oxide synthase (nNOS) from the DGC and sarcolemma to the sarcoplasm is a novel signaling event that stimulates proteolysis, and thus atrophy. New clues can be gleaned from Duchenne and many of the limb-girdle muscular dystrophies, pathologies involving mutations to DGC and other sarcolemmal proteins. Because MDs and microgravity both increases susceptibility of muscles to damage, oxidative stress, and pro-inflammatory signaling, a shared root cause is suspected: disruption of the subsarcolemmal cytoskeleton. New studies and Preliminary Data indicate the importance of NAD(P)H oxidase (NOX), a sarcolemmal source of oxidative stress, and caveolin-3 in stimulating myopathy in muscular dystrophy models. NOX may be activated upstream by angiotensin II and c-abl, a tyrosine kinase. Further, our pilot data show that unloading-induced alterations in caveolin-3 and nNOS are redox dependent and linked to atrophy and shift in muscle fiber-type. Our central hypothesis is that oxidative stress produced by NOX pathways and mitochondria directly contribute to a sarcolemmal loss of nNOS during mechanical unloading, enhancing proteolysis and suppressing protein synthesis. The rodent hindlimb unloading model and novel fractional synthesis and degradation will be used to test our hypotheses. Specific Aim 1 will Identify NOX-dependent signaling on nNOS translocation and muscle fiber atrophy with hindlimb unloading. Specific Aim 2 will determine directly the role of mitochondrial and cytosolic oxidative stress in movement of nNOS away from the muscle cell membrane and disruption of DGC with mechanical unloading. In short, our primary objective is to identify the trigger by which skeletal muscles switch from being an anabolic to a catabolic machine during spaceflight, causing muscle atrophy and weakness. We propose that oxidative stress produced by a NOX pathway and mitochondria move the master switch protein (nNOS) away from the cell membrane, which slows protein synthesis and accelerates breakdown of protein. Our study would directly address high-priority gaps in Recapturing a Future for Space Exploration including the role of reactive oxygen species in protein balance and maintaining the slow contractile phenotype. Our grant proposal would also forward our understanding of (1) alterations in the subsarcolemmal cytoskeleton and membrane domains, (2) sensing and transduction of loading and gravity, (3) protein turnover, and (4) regulatory mechanisms that govern alterations in skeletal muscle, all programmatic emphases and goals of the current Space Biology NRA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Bedrest, casting, and spaceflight cause significant skeletal muscle atrophy and weakness as a result of disuse. Prolonged spaceflight presents great challenges to the physical performance and health of astronauts as a result of microgravity, or reduced gravitational forces (i.e., loads). Mechanical unloading and extreme disuse during spaceflight produce rapid and dramatic skeletal muscle and weakness. Prolonged disuse also predisposes skeletal muscle to damage when landing on Mars, in an EVA (extra-vehicular activity) or returning to Earth. Furthermore, chronic disuse also contributes to and exacerbates underlying chronic pathology, including insulin resistance and cardiovascular disease. As a result crew health, safety, and completion of mission objectives are placed in jeopardy. While the pathways that cause atrophy and weakness due to disuse remain uncertain, proteins that sense changes in forces that skeletal muscles must produce may be the key. One of these proteins is nitric oxide synthase (nNOS), which is positioned near the muscle cell membrane. During spaceflight as well as aging and some types of muscular dystrophy, nNOS appears to be transported away from the cell membrane where it transforms from a muscle building to a muscle wasting agent. We will focus on oxidative stress, produced by excess free radicals produced by proteins near the cell membrane (ex. NOX) and mitochondria and how they affect nNOS transport as well as skeletal muscle atrophy. Understanding how these key membrane proteins are regulated will not only aid us in our treatment of disuse-related atrophy, but also will allow development of new therapeutics combating sedentary lifestyle and possible application to aging, Type 2 diabetes, and muscular dystrophies. Indeed, changes that are altered during spaceflight and ground models of spaceflight also appear to change with aging and diseases that cause skeletal muscle wasting and weakness.
	We moved in September and October 2012 to newly renovated facilities in Heldenfels Hall. While this resulted in 2 months downtime, we have now outstanding facilities, including a BSL-2 room with a Biosafety cabinet and 2 core lab fume hoods. NASA projects will be primarily performed in the new facilities as well as core animal and technology facilities on the Texas A&M campus. We have made significant progress on the NASA Space Biology grant, and that report is below: Recent evidence and our new data suggest that oxidative stress produced via mitochondrial and membrane protein (ex. NADPH oxidase or NOX) are sources of oxidative stress that may participate in muscle atrophy and weakness that occurs with spaceflight. We are using a targeted approach to test the hypothesis that oxidative stress contributes to the transport of nNOS away from the cell membrane. We are using highly specific agents that include (a) novel mimetics of the enzymes superoxide dismutase and catalase (EUK-134) and (b) a mitochondrial-specific antioxidant SS-31. EUK-134 is non-toxic, removes free radicals producing water and oxygen, and does not affect natural antioxidants in the body. Initial experiments used EUK-134 to study the early responses to spaceflight using a ground model with rats. EUK-134 reduced muscle atrophy significantly during short-term spaceflight simulation. EUK-134 also slowed a conversion of slow-twitch to fast-twitch muscle fibers that occurs with spaceflight. Levels of NOX and oxidative stress were reduced by EUK-134 during simulated spaceflight as well. When EUK-134 reduced oxidative stress, it also preserved the location of nNOS near the cell membrane. When nNOS is transported away from the cell membrane, it appears to cause atrophy by turning on proteins that trigger digestion of muscle proteins involved in contractions. One of those proteins activated by movement of nNOS is called FOXO. Our initial experiments found that EUK-134 reduces FOXO activation and therefore slows the muscle atrophy

Task Progress:	that occurs. Therefore, a targeted antioxidant drug like EUK-134 could slow down muscle atrophy that occurs during spaceflight. While we don't understand yet how nNOS is transported away from the muscle cell membrane during spaceflight, one protein that may contribute to a loss of nNOS in Duchenne muscular dystrophy is called caveolin-3. We found that EUK-134 reduced the increase in caveolin-3 that occurred during spaceflight simulation. This means that changes in caveolin-3 and nNOS are controlled by oxidative stress during spaceflight. The emotional stress and loss of gravitational forces could impair glucose tolerance and uptake of glucose during spaceflight. This may increase oxidative stress and contribute to the muscle atrophy we see in astronauts during spaceflight. In order for glucose to be taken in by a muscle cell and transport protein GLUT-4 is attached to glucose to allow entry into the cell. We conducted a study lasting for 7 days of simulated spaceflight. A week of spaceflight reduced skeletal muscle mass and muscle fiber thickness. Intervention with EUK-134 during spaceflight attenuated muscle atrophy. However, we found that GLUT-4 levels at the muscle membrane were higher, rather than lower during our ground simulation. The metabolic regulator AMPK was also higher rather than lower, normalized by EUK-134. These new findings indicate that skeletal muscles are able to compensate their metabolism with spaceflight and are even more capable of taking up glucose for fuel. The data are also inconsistent with the notion that stress hormones such as cortisol (which oppose insulin) released during reduced gravity are enough to cause insulin resistance and impairment of glucose uptake (diabetes-like). We used 2 methods of fluorescence and histological staining to show that oxidative stress contributes to loss of nNOS from the muscle cell membrane after one-week of simulated spaceflight. Both methods clearly showed that nNOS had been transported away from the muscle cell membrane, and this effect was blunted by administration of EUK-134. We also found that NOX was localized in "hot spots" near the cell membrane, and effect also attenuated by EUK-134 therapy. We believe that NOX is acting as an amplifier of oxidative stress, perhaps working with mitochondria located near the cell membrane. Muscle damage can be caused by stressful situations including excessive exercise, prolonged disuse, and disease. Heat shock proteins such as HSP70 can reduce muscle atrophy. Combinations of anti-inflammatory drugs, antioxidants, and nutrients have been shown to prevent or reduce damage and muscle wasting. We were interested in testing the effects of geranylgeranylacetone (GGA), a lycopene precursor, in protecting against skeletal muscle damage caused by disuse. Further, we want to understand the extent in which GGA induces the expression of heat shock protein 70 (HSP70) in muscles from middle-age Fischer-344 rats. Heat shock proteins are especially important in muscle mass and integrity because they have been proven to exhibit cytoprotective function including inhibition of apoptosis, and the transcription factor FOXO. To address our first question, we used young and middle aged Fischer rats in a hindlimb unloading experiment, as it is a common model for disuse. We unloaded rats for two days and injected them with GGA daily beginning 24 hours prior to unloading. After the experiment, we dissected the soleus and conducted an assessment of internal/central nuclei assessment and muscle fiber cross-sectional area. Our results from the first part of our study indicate that GGA does play a significant role in protecting and restoring skeletal muscle health. Specifically, our data showed a significant difference in the number of internal and central nuclei between treatment groups, which serve as a marker for damage and repair. Our future research will continue with this study to test GGA's ability to protect muscle cell protection and regeneration, and its relationship with HSP70 induction. Sometimes nitric oxide produced by nNOS or another type of NOS could react with free radicals produced by NOX to form peroxynitrite, which is highly dangerous. We used a stain for 3-nitrotyrosine (3NT) as a marker. We found that 3NT was indeed greater with 7 days of simulated spaceflight, particularly near the membrane. Recently, it has been shown that the pro-inflammatory iNOS may participate in unloading-induced atrophy, although how this occurs is unsure. As astronauts are usually middle-age (40-55) and are growing older on average, we sought to determine whether oxidative stress was responsible for the reduction in nNOS. We have shown previously that aging cause a reduction in nNOS including nNOS localized at the sarcolemma. Lifelong exercise or caloric restriction reduced the amount nNOS loss as well as muscle fiber atrophy. We have recently found that (1) oxidative stress contributes to loss of nNOS from the sarcolemma with disuse and DMD, and (2) reduction of nNOS in aging skeletal muscle is reduced by exercise training while oxidative stress is decreased. We postulated that oxidative stress may be causal in the loss of nNOS from the sarcolemmal in aging skeletal muscle. To test this we used EUK-134. Old Fischer-344 rats were administered EUK-134 for 28 days. Young and middle-age F344 rats were used as additional controls. We dissected and froze the plantaris in isopentane cooled in liquid nitrogen. nNOS was localized using immunohistochemistry with a tyramide signaling amplification. We found that nNOS-positive staining decreased in the plantaris from old rats compared with young and mature rats. EUK-134 provided partial protection from the loss of sarcolemmal nNOS. Transforming growth factor-ß (TGF-ß) was elevated with age, but attenuated with EUK-134. Another protein that can cause muscle wasting (MMP-9) saw an elevation with age that was suppressed by EUK-134. To our knowledge this study is the first evidence that age-related loss of nNOS from muscle sarcolemma is dependent on oxidative stress. Future research will focus on (a) the mechanisms of nNOS protection and (b) protection against damage with aging.
Bibliography Type:	Description: (Last Updated: 11/13/2025)
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